

MedLedger Health Exchange - Structured Architecture

System: MedLedger Clinical Exchange

Classification: Restricted

Deployment: Hybrid

Regulatory Scope: PIPEDA, ISO 27001, NIST

Trust Boundary: Patient and Partner Edge

Contains:

- Identity Gateway [process]

Trust Boundary: Clinical Operations Zone

Contains:

- Clinical Integration Bus [process]
- Break-Glass Service [process]
- Vendor Support Bastion [process]
- Imaging AI Gateway [process]

Trust Boundary: Restricted Clinical Data Zone

Contains:

- Electronic Health Record Store [data_store]
- PACS Archive [data_store]
- Lab Results Repository [data_store]

Data Flows:

- Patient Portal User -> Identity Gateway: patient login and records access
- Identity Gateway -> Clinical Integration Bus: authenticated patient session
- Clinical Integration Bus -> Electronic Health Record Store: patient chart query
- Clinical Integration Bus -> PACS Archive: imaging retrieval
- Clinical Integration Bus -> Lab Results Repository: laboratory results query
- Insurer API -> Clinical Integration Bus: eligibility and claims status
- Break-Glass Service -> Electronic Health Record Store: emergency override query
- Vendor Support Technician -> Vendor Support Bastion: privileged vendor session
- Vendor Support Bastion -> PACS Archive: diagnostics session
- Clinical Integration Bus -> Imaging AI Gateway: imaging inference request

